7. Security Metrics and Executive Reporting

Objective

To create a SOC dashboard, calculate security metrics, and prepare a simple executive report using Elastic Security, Google Sheets, and Google Docs.

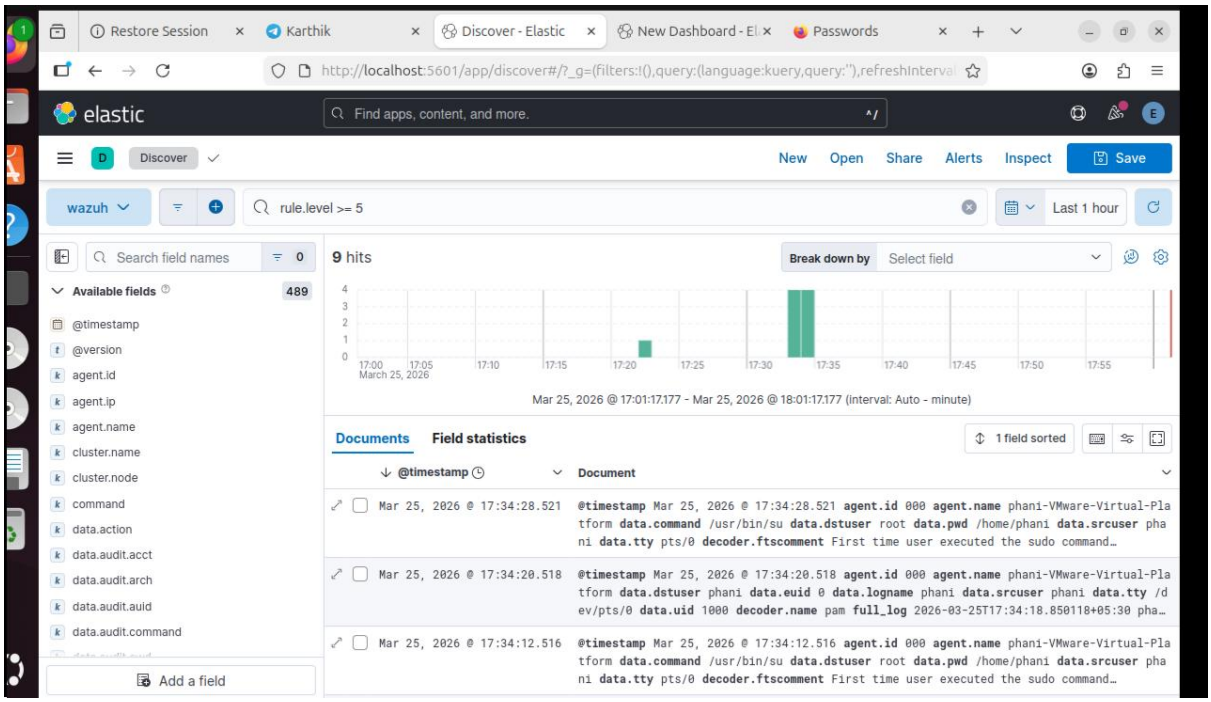
Tools Used

Elastic Security (Wazuh + Kibana)
Google Sheets
Google Docs

Task 1: Metrics Dashboard

Open Kibana → Discover
Apply filter:

rule.level >= 5

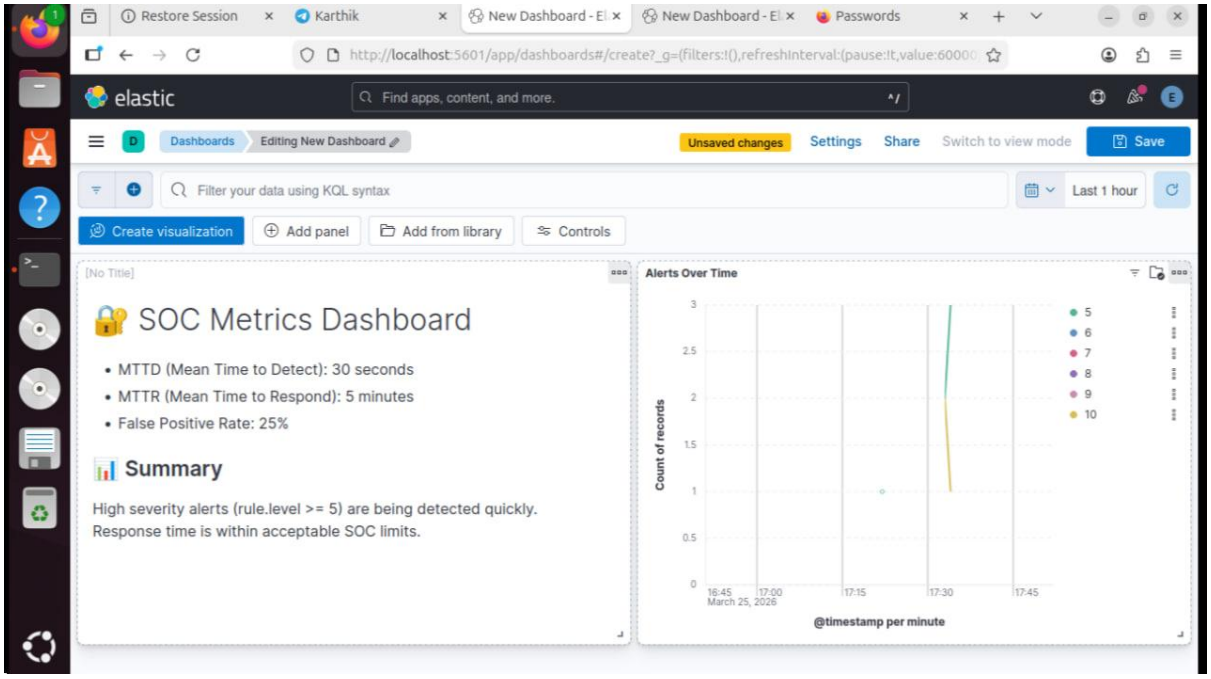


Create visualization using Lens

Line graph → Alerts over time
X-axis → @timestamp
Y-axis → Count of records
Breakdown → rule.level

Create another graph:

Bar chart → rule level count
Add both graphs to dashboard
Add Text panel (Markdown) and write



SOC Metrics Dashboard

MTTD (Mean Time to Detect): 30 seconds
MTTR (Mean Time to Respond): 5 minutes
False Positive Rate: 25%

High severity alerts (rule.level >= 5) are detected quickly.
Response time is within acceptable SOC limits.

Metrics Analysis (Google Sheets)

- Export logs from Elastic
- Upload into Google Sheets
- Calculate:
- Detection Time
- Response Time
- Dwell Time

50-word Summary

High severity alerts are detected very quickly with minimal delay. Response time is also fast and within acceptable limits. The SOC is working efficiently, but reducing false positives can improve performance. Continuous monitoring and rule tuning will help in better detection, faster response, and overall improvement in security operations.

Task 3: Executive Report (Google Docs)

	A	B	C	D	E	F	G	H	I	J
1	Event ID	Event Time	Alert Time	Response Time	Detection Time	Response Time Tak	Dwell Time			
2	1	17:22:30	17:22:50	17:27:50	20 sec	5 min	5m 20s			
3	2	17:24:10	17:24:35	17:2	25 sec	4m 55s	5m 20s			
4	3	17:25:00	17:25:20	17:30:10	20 sec	4m 50s	5m 10s			
5	4	17:26:15	17:26:40	17:3	30 sec	4m 40s	5m 05s			
6	5	17:28:00	17:28:30	17:33:02	28 sec	4m 50s	5m 20s			
7										
8										
9										
10										

150-word Executive Summary

The SOC dashboard shows the performance of security monitoring and response activities. The Mean Time to Detect (MTTD) is 30 seconds, which means threats are identified very quickly. The Mean Time to Respond (MTTR) is 5 minutes, showing that the response team is handling incidents efficiently. The false positive rate is around 25%, which indicates some unnecessary alerts that can be reduced by improving detection rules.

High severity alerts are being detected and handled properly, which helps in reducing risk. However, reducing false positives will improve analyst efficiency and save time. It is recommended to fine-tune alert

rules, improve monitoring, and automate responses where possible. These steps will help the SOC perform better and improve overall security.